

Master Deployment Guide: 24/7 Ops Console

COMPREHENSIVE REPLICATION INSTRUCTIONS (V4.0 - FINAL)

Phase 1: Hardware & BIOS Configuration

Ensure the Beelink MiniPC restores operation immediately after a power outage.

- **Auto Power-On:** Boot and tap **Del**. Go to **Advanced > Power Management**. Set **Restore AC Power Loss** to **Power On** (or **S0 State**). Save & Exit (F4).

Phase 2: OS Preparation (Windows 11)

1. Passwordless Autologon

- **Registry Tweak:** Open **regedit**. Navigate to: **HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion>PasswordLess\Device**. Change **DevicePasswordLessBuildVersion** to **0**.
- **netplwiz:** Press **Win+R**, type **netplwiz**. Uncheck "Users must enter a user name and password...". Click Apply and enter the account password to save credentials.

2. Power & Sleep Management (Anti-Modern Standby)

- **Disable S0 Sleep:** In **regedit**, go to **HKLM\System\CurrentControlSet\Control\Power**. Create a DWORD (32-bit) **PlatformAoAcOverride** and set it to **0**.
- **Global Sleep Settings:** Set Screen and Sleep to **Never** in System Settings.
- **USB Selective Suspend:** Control Panel > Power Options > Change advanced power settings > USB settings > Set **USB selective suspend** to **Disabled**.

Phase 3: VS Code Profile & Environment

1. The "AutoDebug" Profile

- In VS Code, click **Gear Icon > Profiles > Create Profile**. Name it **AutoDebug**.
- Ensure the **AU** badge is visible in the bottom-left corner.

2. Extension & Autorun Config

Install **Python** and **Auto Run Command**. Open Profile Settings (JSON) and add:

```
{
  "auto-run-command.rules": [
    { "command": "workbench.action.debug.start", "event": "onStartupFinished" }
  ],
  "workbench.startupEditor": "none"
}
```

Phase 4: Project Configuration & Launch

1. Defining Arguments (launch.json)

Open `.vscode/launch.json` and configure your arguments:

```
{
  "name": "Gen20psConsole_Debug",
  "type": "debugpy",
  "request": "launch",
  "program": "${workspaceFolder}/MasterControl.py",
  "console": "integratedTerminal",
  "args": ["--port", "COM3", "--verbose"]
}
```

2. Startup Shortcut

Press `Win+R`, type `shell:startup`. Create a shortcut with the following target:

```
"C:\Path\To\Code.exe" "C:\Path\To\Project.code-workspace" --new-window --profile "AutoDebug"
```

Appendix A: Python Keepalive Fail-safe

Add this to the start of `MasterControl.py` to prevent sleep if Registry settings are reverted by Windows Update:

```
import ctypes
def prevent_sleep():
    # ES_CONTINUOUS | ES_SYSTEM_REQUIRED
    ctypes.windll.kernel32.SetThreadExecutionState(0x80000000 | 0x00000001)
```

```
prevent_sleep()
```

Appendix B: Paranoia-Level USB Control

For critical equipment control, disable power management per device:

1. Open **Device Manager**.
2. Right-click **USB Root Hub** > **Properties**.
3. **Power Management** tab > Uncheck *"Allow the computer to turn off this device..."*.